DenseKD: Dense Knowledge Distillation by Exploiting Region and Sample Importance

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***Abstract*— Knowledge distillation (KD) can compress deep neural networks (DNNs) by transferring the knowledge of the redundant teacher model to the resource-friendly student model, where cross-layer KD (CKD) conducts KD between each stage of students and the multiple stages of teachers. However, previous CKD schemes select the coarse-grained stagewise features of teachers to teach students, leading to improper channel alignment. Also, most of these methods conduct uniform distillation for all the knowledge, limiting students to focus more on important knowledge. To address these problems, we propose a dense KD (DenseKD) in this article, dubbed as DenseKD. First, to achieve more accurate feature alignment in CKD, we construct the learnable dense architecture to make each channel of student flexibly capture more diverse channelwise features from teacher. Moreover, we introduce region importance to investigate the region's guiding potential, it distinguishes the influence of different regions by the variation of representations of teacher models. In addition, to make students pay more attention to useful samples in KD, we calculate sample importance by the loss of teacher models. Consistent improvements over state-of-the-art approaches are observed in experiments on multiple vision tasks. For example, in the classification task, DenseKD achieves 72.30% accuracy of ResNet-20 on CIFAR-100, which is higher than the results of previous CKD methods. In addition, in the object detection task, DenseKD gains 2.84% mean average precision (mAP) improvements of Faster R-CNN with ResNet-18 against vanilla KD.**

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Code is available on-line at https://github.com/zhnxjtu/DenseKD.

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***Index Terms*— Deep neural networks (DNNs), knowledge distillation (KD), model compression, region importance, sample importance.**

I. Introduction

Deep neural network (DNN) shows powerful capacity in a large number of vision tasks, such as, classification [1], [2], object detection [3], [4], and segmentation [5], [6]. However, redundant parameters and repeated operations limit the application of DNNs in resource-constrained hardware devices [7], [8]. Therefore, several model compression approaches are proposed to produce the lightweight model, such as quantization [9], [10], [11], compact model design [12], [13], [14], low-rank decomposition [15], [16], [17], [18], network pruning [19], [20], and knowledge distillation (KD) [21], [22], [23], [24], [25]. In this article, we mainly focus on KD, especially, feature-based KD [26], because it can be efficiently applied to existing architectures, avoiding the cumbersome search for lightweight network architectures, and can be combined with other compression schemes to further improve the accuracy of compact models [8].

In the field of model compression, KD has been validated as one of the potential techniques to gain a smaller sized lightweight model (i.e., student) with the supervision of pre-trained larger sized model (i.e., teacher). Bucilua et al. [27] first leverage a more complex model to guide the learning procedure of a simpler model. Hinton et al. [21] further popularize this method, and name it as KD. The core idea of KD is to mimic the teacher network with a compact student network. Vanilla KD method [21] regards the softened logit of the teacher as an extra supervised signal to train the student. Although the above scheme has ability to transfer the knowledge from the teacher network to student one, the limited supervised information leads to the unsatisfactory performance of the student model.

Since the features in DNNs contain more abstract representations, feature-based distillation methods are further proposed to achieve the high-performance student model. Romero et al. [26] introduce "hint" to match the features of the student–teacher pair. Instead of mimicking the output features of the teacher, Zagoruyko and Komodakis [28] calculate the attention map of each stage for knowledge transfer. The above feature-based distillation methods construct a feature transfer bridge between the same stage's features of teacher and student networks. Different from the above methods, several recent methods focus on cross-layer distillation, which uses the multiple stages' features of the teacher to supervise single-stage's learning of students. For example, Chen et al. [29] leverage

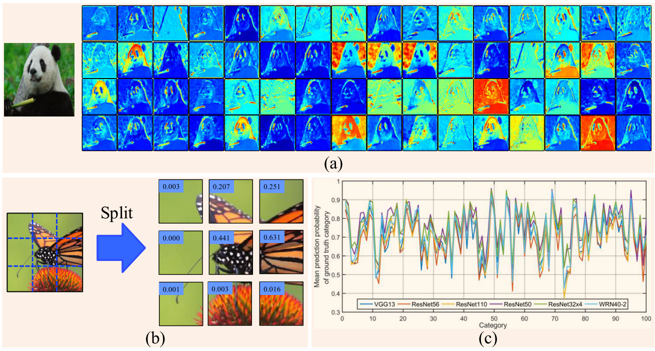


Fig. 1. Illustration of phenomena in DNNs. (a) Feature map of each channel generated by the first block of ResNet-18. (b) Prediction of local regions by ResNet-34, the number below each region represents the predicted value (output of Softmax) for the corresponding local region. (c) Prediction results of images in different categories, where x- and y-axes represent the index of category and the mean of prediction probability of ground-truth category, respectively.

the self-attention mechanism [30] to calculate the weight of features of each teacher stage, and then encourage single-layer feature of student to mimic the weighted multilayer features of teacher. In [31], the shallower features of teacher are transferred to the current layer of student, which is called the knowledge review mechanism.

1) *Coarse-Grained Distillation Position Selection:* Previous cross-layer distillation methods [29], [31] conduct stage-wise selection to match multiple student features with single teacher features. However, as shown in Fig. 1(a), the channelwise features of each stage in DNNs are diverse, they tend to flexibly capture the channelwise features of different stages in teacher models. The stage-wise selection performs the same operation on all the features in the same stage; then, the knowledge transfer of any channel is constrained by other channels in the same stage, which causes inaccurate feature alignment.
2) *Uniform Distillation for All Regions:* Some distillation methods [29], [32] ignore the influence of different local regions in knowledge transfer. However, as shown in Fig. 1(b), different regions in the same sample (or feature) play diverse roles for object recognition; hence, they tend to have different potential to guide the student learning, and the uniform distillation for all regions makes students fail to capture the differential supervision of different regions.
3) *Uniform Distillation for All Samples:* For KD on classification tasks, a large number of distillation methods [29], [31], [32] regard that all the samples provide the same influence on feature transfer. However, as shown in Fig. 1(c), the DNNs show different capacities to classify samples with different categories, which indicates that the networks have different feature extraction abilities for different samples. Hence, different samples in dataset have diverse guiding potential for distillation. Transferring all samples in a uniform manner limits the student networks to learn more from important samples.

To address the above problems, in this article, we propose a fine-grained dense cross-layer knowledge distillation (DenseKD), and investigate the region and sample importance in the feature alignment process. First, to achieve more accurate and sufficient cross-layer distillation, we construct the learnable dense architecture to adaptively select channelwise features for knowledge transfer. The dense architecture consists of three modules: refining, fusion, and transformation modules. The refining module is used to reduce the output dimension to accelerate the transfer process, and it selects and purifies the channelwise feature in each stage. The second one is leveraged to select and fuse the features of all the stages, and then the transformation module transforms the selected channelwise features to fit the teacher features at each stage. In addition, to prevent uniform distillation for different regions in each feature, we introduce region importance to demonstrate the guiding capacity of different local feature regions in distillation. The region's importance is calculated by the difference between the abstract features of the local region and those of the original image without relying on the ground truth. In addition, to alleviate the uniform distillation for different samples in the dataset, we further introduce sample importance to illustrate the guiding capacity of different samples, and we propose to apply the loss of each sample in the teacher network to calculate the importance of corresponding sample.

In summary, the contribution of this article is fourfold.

1) A learnable dense connection is proposed to accurately execute the cross-layer distillation in an adaptive and fine-grained scheme.
2) The region importance calculated by the variation of teacher features is used to investigate the guiding potential of different regions for distillation.
3) The sample importance calculated by the teacher loss is leveraged to illustrate the guiding potential of different samples for distillation.
4) Extensive experiments on multiple vision tasks, including, classification, object detection, and instance segmentation, validate that DenseKD brings improvements over the state-of-the-art approaches. In addition, abundant ablation studies further illustrate the effectiveness of each component in DenseKD.

The rest of this article is organized as follows. In Section II, we introduce related works and demonstrate the discrepancy between DenseKD with other advanced works. Moreover in Section III, we first define the symbols, and then illustrate the details of each component in the proposed method. After that, in Section IV, we compress different DNNs on several vision tasks to illustrate the preferable gains of our method. In Section V, the ablation study further demonstrates the effectiveness of DenseKD. Finally, we conclude in Section VI.

II. Related Works

A. Feature-Based KD

As one of the important components of DNNs, the intermediate feature is the powerful information for distillation. Feature-based KD conducts the feature alignment between the combination of teacher and student models for knowledge transfer. Romero et al. [26] first leverage the features of complex model to train a deeper and thinner model, they introduce "hint" into teacher, and align the output of hint with the projected features of the student. Instead of using actual representation of features, Passalis and Tefas [33] transfer the probability distribution of features to student model. Moreover, Tian et al. [34] combine the KD with contrastive learning, they distill more important features by the family of contrastive objectives. CAT-KD [35] leverages the class activation map to indicate the discriminative regions of features, and then conducts KD on these selected regions. To minimize channelwise feature redundancy for KD, DDCD [36] employs a channelwise contrastive objective that reduces the distance of corresponding channels while increasing the distance of noncorresponding channels. These methods can be called single-layer KD (SKD) as they align the features of a single layer.

In addition, a larger number of methods transfer the features of multiple layers, which is called multiple-layer KD (MKD). Zagoruyko and Komodakis [28] calculate the norm of features to generate attention map as the transferred knowledge. Ahn et al. [37] distill the knowledge by maximizing the mutual information between the teacher and the student networks. Moreover, Heo et al. [38] propose the margin rectified linear unit (ReLU), novel distillation feature position, and partial $L2$ distance function to skip useless information in KD.

Instead of aligning the same-level features of teacher and student networks, recent cross-layer KD (CKD) methods leverage the features of multiple teacher stages to guide the training of each student stage (i.e., the features of each teacher stage can teach multiple student stages). Chen et al. [29] illustrate the influence of features at different teacher stages by self-attention mechanism [30], and transfer the weighted features of multiple teacher stages to each student stage. In addition, Ji et al. [32] also leverage the self-attention mechanism to propose a feature linking method for multistage feature alignment. In addition, the knowledge review mechanism is introduced in [31]. The authors manually specified the teacher–student stage pair, and move the shallower features of teacher to the current stage of the student.

B. KD for Detection

KD can not only compress the models for classification tasks but it also shows powerful potential in object detection tasks [39], [40], [41], [42]. To generalize KD to the detection task, a large number of methods attempt to balance the influence of different local instances (i.e., positive and negative instances). For example, Chen et al. [39] distill the knowledge in backbone and region proposal network (RPN) by "hint" and vanilla KD, respectively. Specifically, the weighted soften logit loss is used to keep the balance between foreground and background in distillation. In addition, Li et al. [40] distill a certain number of proposal regions (generated by RPN) instead of global features, leading a balance transfer for foreground and background regions. To pay more attention to the positive instances, Wang et al. [41] present to transfer the knowledge of object regions and near object regions, where the regions are selected by the intersection over union (IoU) of predicted anchors w.r.t. ground-truth bounding boxes. Furthermore, Dai et al. [43] compare the predictions of teacher and student models to select the discriminative instances for KD. FreeKD [44] conducts distillation in the frequency domain, and it introduces a learnable frequency prompt module in the teacher model to select the pixel of interest in the frequency domain. Moreover, Kang et al. [45] build the query–key pair to evaluate the contribution of different local regions, and then select the important regions for distillation.

C. Discussion

Most of feature-based distillation methods, especially CKD, bring preferable performances, yet the coarse-grained distillation position matching, and uniform distillation scheme for all the regions of features and all the samples in the dataset limit more accurate feature alignment. This article focuses on the more fine-grained CKD, involving region and sample importance during feature matching. The proposed method is completely orthogonal to previous feature-based distillation, since it conducts a fine-grained channelwise selection to create the path for knowledge transfer between the teacher–student pairs. Also, unlike transferring attention map in AT [28], we transfer the masked feature maps when investigating region importance. In addition, the region importance is calculated without leveraging the ground truth, which is different from [46] and [47]. In addition, DenseKD also has orthogonality with previous KD methods for object detection, because: 1) the sample importance in DenseKD is used to distinguish the guiding capacity between different global images, rather than that of different local instances captured by the bounding boxes (or anchors) and 2) also, the region importance in our method is adopted to illustrate the importance of nonoverlapped local features in the backbone model, rather than that of the positive and negative instances.

III. Methodology

In this section, we first formally introduce the symbols and notations of KD. After that, the framework of DenseKD is introduced in Sections III-B–III-D, which includes: 1) as shown in Fig. 2(a), we propose a dense connection architecture to conduct a fine-grained CKD; 2) as shown in Fig. 2(b) and (c), we investigate region importance to illustrate the guiding potential of different regions in each feature; and 3) as shown in Fig. 2(b) and (d), the sample importance of different sample is also calculated to explore the guiding potential of different samples in dataset.

A. Preliminary

We first formally introduce the symbols and notations of KD in this section. We leverage $\boldsymbol{\chi}^N = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ to represent the dataset consisting of N samples, where $\mathbf{x}_i$ and $\mathbf{y}_i$ represent the sample and corresponding label, respectively, and the purpose of KD is to obtain the simple student model by reusing the pretrained teacher model and same dataset. We use $\mathbf{F}_{l_t}^t \in \mathbb{R}^{b\times c_{l_t}\times h_{l_t}\times w_{l_t}}$ and $\mathbf{F}_{l_s}^s \in \mathbb{R}^{b\times c_{l_s}\times h_{l_s}\times w_{l_s}}$ to represent the features of l_tth teacher stage and l_sth student stage, respectively. For teacher features $\mathbf{F}_{l_t}^t$, b represents the mini-batch size, and c_{l_t} and $h_{l_t} \times w_{l_t}$ represent the number of channel and spatial size, respectively. $l_t \in [1, L_t]$, where L_t represents the number of teacher stages for distillation. All these symbols corresponding to student features ($\mathbf{F}_{l_s}^s$) have similar meanings. $\mathbf{F}_{l_t}^t(\mathbf{x}_i) \in \mathbb{R}^{c_{l_t}\times h_{l_t}\times w_{l_t}}$ and $\mathbf{F}_{l_s}^s(\mathbf{x}_i) \in \mathbb{R}^{c_{l_s}\times h_{l_s}\times w_{l_s}}$ represent the features of teacher and student generated by sample $\mathbf{x}_i$, respectively. For distillation between homogeneous teacher and student (e.g., the combination of "ResNet-56 and ResNet-20"), $L_t = L_s$. For distillation between heterogeneous groups (e.g., the combination of "VGG-13 and MobileNetV2"), $L_t \neq L_s$.

In this article, we mainly focus on feature-based KD, and the loss of vanilla feature-based KD can be formulated as follows:

$$\mathcal{L}_{\mathrm{KD}_i} = \sum_{l=1}^{L} \mathcal{D}\left(\mathcal{T}_l^t\left(\mathbf{F}_l^t(\mathbf{x}_i)\right), \mathcal{T}_l^s\left(\mathbf{F}_l^s(\mathbf{x}_i)\right)\right) \tag{1}$$

where $\mathcal{T}^t$ and $\mathcal{T}^s$ represent the transformation of features of teacher and student, respectively, such as attention map [28]

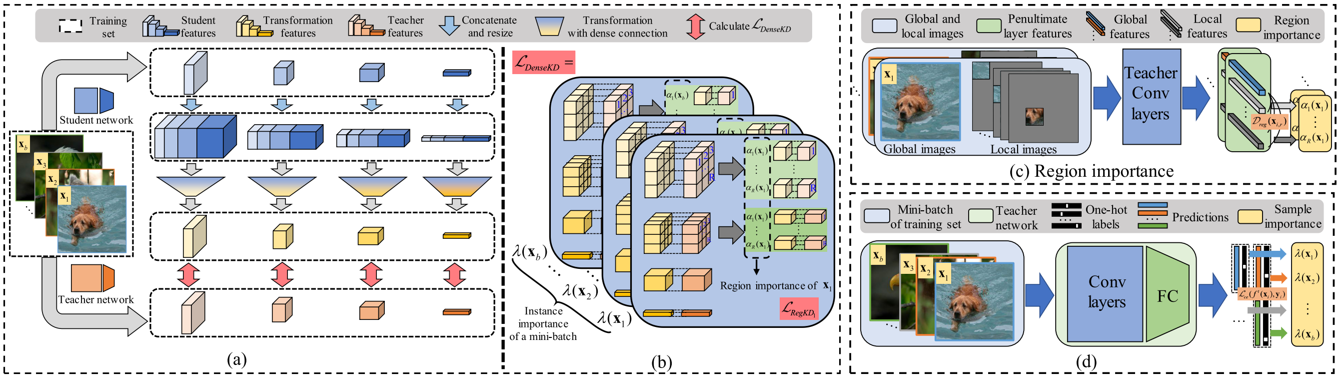


Fig. 2. Framework of the proposed method. (a) Overall framework of DenseKD. First, the features of all student stages are concatenated and resized. After that, the dense connection is leveraged to generate the transformation features. Finally, the region and sample importance are investigated for knowledge transfer. (b) Details of knowledge transfer with region and sample importance. (c) and (d) Calculation of region importance and sample importance, respectively.

and factor [48]. $\mathcal{D}$ measures the distance between the knowledge of teacher and student. For SKD, $l = L = L_t$ or L_s. For MKD, $L = \min(L_t, L_s)$.

B. Distillation With Dense Connection

Equation (1) helps the student model capture the features of a teacher at the same stage, while the features of the same stage can be mismatched, which tends to provide a suboptimal student network. To further get high-performance student model, recent CKD methods [29], [32] select the multistage features of teacher to supervise each-stage learning of the student. The loss function of CKD can be expressed as follows:

$$\mathcal{L}_{\text{CKD}_i} = \sum_{l_s=1}^{L_s} \sum_{l_t=1}^{L_t} \mathcal{D}\big(\mathcal{T}_{l_t}^t\big(\mathbf{F}_{l_t}^t(\mathbf{x}_i)\big), \mathcal{T}_{l_s}^s\big(\mathbf{F}_{l_s}^s(\mathbf{x}_i)\big)\big). \tag{2}$$

In (2), the features of each student stage are matched with the features of the 1st to L_tth teacher stages. We change the order of summation, and rewrite (2) as follows:

$$\mathcal{L}_{\text{CKD}_i} = \sum_{l_t=1}^{L_t} \sum_{l_s=1}^{L_s} \mathcal{D}\big(\mathcal{T}_{l_t}^t\big(\mathbf{F}_{l_t}^t(\mathbf{x}_i)\big), \mathcal{T}_{l_s}^s\big(\mathbf{F}_{l_s}^s(\mathbf{x}_i)\big)\big). \tag{3}$$

Equation (3) indicates that the features of each teacher stage can guide the multistage learning of the student. In (3), to find the position for distillation, previous methods conduct a coarse-grained stagewise selection to match the features of multiple student stage with those of each teacher stage. However, the channelwise features in the same stage of student can be supervised by channelwise features in different stages of teacher, thus previous stagewise selection methods ignore the diversity of features in the same stage, leading to improper distillation path between teacher–student pairs. To alleviate this issue, we construct the learnable dense architecture to match multiple stages of student with each stage of teacher in a fine-grained manner.

The constructed connection needs to select, fuse, and transform the features of each channel at different student stages to match the l_tth layer's teacher features; hence, the input and output size of dense connection are $\sum_{l_s=1}^{L_s} c_{l_s}$ and c_{l_t}, respectively. As shown in Fig. 3(a), we leverage one layer with 1×1 convolution to select and fuse student features, the input and output of this layer are both $\sum_{l_s=1}^{L_s} c_{l_s}$, and each channel of output fuses the channelwise features of all student

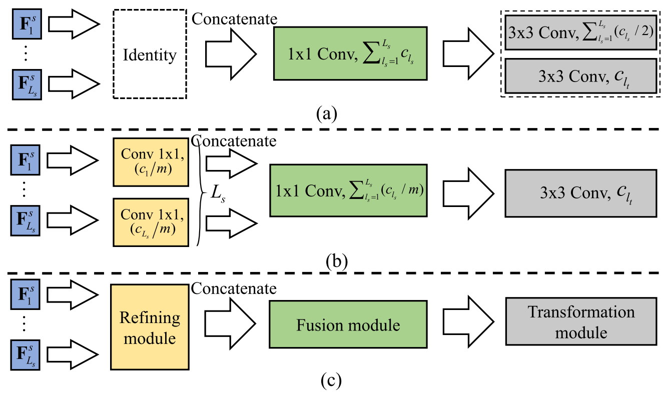


Fig. 3. Architecture of dense connection. (a) and (b) Basic and fast dense connection architectures, respectively. (c) Summary of dense connection.

stages. Furthermore, two layers with 3×3 convolutions are used to transform the features. The basic architecture is helpful for the student model with fewer channels at all stages (e.g., ResNet-20 has 176 channels). However, the model with plenty of channels at all stages brings cumbersome computation and storage. For example, when VGG-13 as the teacher and VGG-8 as the student, the number of channels at all stages are 1920, leading to $1920 \times 1920 \times 1 \times 1$ of parameters in 1×1 layer, which is time-consuming for training.

To reduce the knowledge alignment overhead of this type of distillation combinations, as shown in Fig. 3(b), we insert a layer with 1×1 convolution for each-stage feature before selection and fusion layer to reduce the number of output channels at each stage, where the number of input and output channels of stage l_t are c_{l_s} and c_{l_s}/m, respectively, and m represents the reduction factor. In this case, the parameters before transformation layer becomes $((1/m) \cdot (\sum_{l_s=1}^{L_s} c_{l_s}^2/(\sum_{l_s=1}^{L_s} c_{l_s})^2) + (1/m^2))$ of basic architecture. For example, compared with the basic dense connection, when using fast one with $m = 4$ to select the distillation position of VGG-8, the parameters before the transformation layer are reduced by half. In addition, we further reduce the computation by using one-layer 3×3 convolution for transformation in the fast dense connection.

As shown in Fig. 3(c), we summarize the dense connection architecture into three components: 1) the *refining module* consists of multiple groups of 1×1 convolutions, and it reduces the output dimension, selects and purifies the features of each student stage; 2) in addition, the *fusion module* contains single layer 1×1 convolution, which is used to select and fuse the features of all the layers; and 3) moreover, the *transformation module* with 3×3 convolution aligns the fused multistage student features with teacher features. The distillation loss with a dense connection can be written as follows:

$$\mathcal{L}_{\mathrm{DKD}_i} = \sum_{l_t=1}^{L_t} \mathcal{D}\left(\mathbf{F}^t_{l_t}(\mathbf{x}_i), \widehat{\mathbf{F}^s_{l_t}(\mathbf{x}_i)}\right) \tag{4}$$

where $\mathcal{T}^s$ represents the learnable dense connection and $\widehat{\mathbf{F}^s_{l_t}(\mathbf{x}_i)} = \mathcal{T}^s_{l_t}(\mathbf{F}^s_1(\mathbf{x}_i); \ldots; \mathbf{F}^s_{L_s}(\mathbf{x}_i))$ represents the knowledge transformed from multiple-stage student features.

C. Distillation by Exploiting Region Importance

Equation (4) investigates the diversity of features in each layer and conducts a fine-grained cross-layer distillation; however, it still transfers all the local regions of features equally (i.e., uniform distillation). As stated in [47], [49], and [50], different regions of features play different roles for classification; thus, they tend to have different guiding potentials for knowledge transfer, and the importance of different regions needs to be investigated in (4). Then, the distillation loss can be represented as follows:

$$\mathcal{L}_{\mathrm{RegKD}_i} = \sum_{l_t=1}^{L}\sum_{r=1}^{R} \alpha_r(\mathbf{x}_i) \cdot \mathcal{D}\left(\mathbf{F}^t_{l_t,r}(\mathbf{x}_i), \widehat{\mathbf{F}^s_{l_t,r}(\mathbf{x}_i)}\right) \tag{5}$$

where $\mathbf{F}^t_{l,r}(\mathbf{x}_i)$ and $\widehat{\mathbf{F}^s_{l_t,r}(\mathbf{x}_i)}$ represent the rth local region of feature maps generated by $\mathbf{x}_i$, $\alpha_r(\mathbf{x}_i)$ represents their importance and R represents the number of regions.

In order to calculate the region importance, we use the teacher model to generate the penultimate layer features of local regions and original image, respectively, and then calculate the differences between local regions' features and image's feature to demonstrate the influence of different regions. The convolutional layer of teacher model is a well-pretrained feature extractor, and the penultimate layer features of original images can be considered as the ground-truth encoding for target judgment. When the penultimate layer features of the local region are much different from the ground-truth encoding, the corresponding local region can hardly affect the network to complete the task; thus, the influence of the region can be reduced in knowledge transfer. On the contrary, when the local region and original image gain the similar penultimate layer features, the corresponding region contains useful features; thus, it is more important. The difference of penultimate layer features can be formulated as follows:

$$\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r}) = \exp\left(\left\langle \overline{\mathbf{F}^t_{L_t}(\mathbf{x}_{i,r})}, \overline{\mathbf{F}^t_{L_t}(\mathbf{x}_i)} \right\rangle\right) \tag{6}$$

where $\overline{\mathbf{F}^t_{L_t}(\mathbf{x}_i)} = \mathbf{F}^t_{L_t}(\mathbf{x}_i)/||\mathbf{F}^t_{L_t}(\mathbf{x}_i)||_2$ represents the unitized penultimate layer features and $\langle\rangle$ represents the inner product. When the rth region contains useful information, $\overline{\mathbf{F}^t_{L_t}(\mathbf{x}_{i,r})}$ and $\overline{\mathbf{F}^t_{L_t}(\mathbf{x}_i)}$ have similar distributions, and then $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ tends to be larger. Conversely, a smaller $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ indicates that the corresponding region has less influence. Since (6) only leverages the backbone of the model (without classification head) to illustrate the region importance, the calculation of region importance does not rely on the ground truth (e.g., one-hot label).

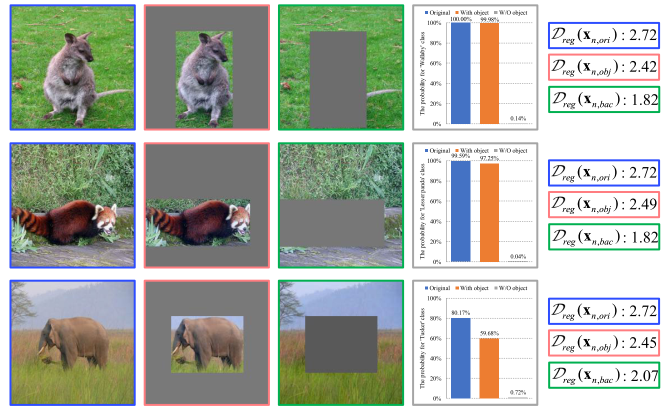


Fig. 4. $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ for object and background regions. 1st, 2nd, and 3rd columns represents the raw images, images with object region, and images with background region, respectively. The fourth column represents the prediction results of different images by ResNet-34. The rightmost column represents $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ of images with different regions.

In Fig. 4, we show the $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ of object and background regions to explain the effectiveness of (6). Intuitively, the object region contains more useful features; thus, its corresponding $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r})$ should be larger. As shown in the rightmost column of Fig. 4, $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{n,\mathrm{obj}})$ is larger than $\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{n,\mathrm{bac}})$, indicating that (6) can effectively help mark the importance of different regions. To unify the range of region importance of different samples, we calculate the region importance by the following formula:

$$\begin{cases} \alpha_r(\mathbf{x}_i) = \dfrac{\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r}) - \min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'}))}{\max(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) - \min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'}))}, \\ \quad \text{if } \max(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) - \min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) > \varepsilon \\ \alpha_r(\mathbf{x}_i) = 1, \\ \quad \text{if } \max(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) - \min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) \le \varepsilon \end{cases} \tag{7}$$

where $\alpha_r(\mathbf{x}_i)$ represents the importance of the rth region, $r' = 1, 2, \ldots, R$. ε represents an infinitesimal positive value, it is leveraged to avoid numerical problems caused by $\max(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'}))$ and $\min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'}))$ being equal or similar (the denominator is equal to 0 or approaches 0). Since different regions have similar region importance when $\max(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'})) = \min(\mathcal{D}_{\mathrm{reg}}(\mathbf{x}_{i,r'}))$, we set $\alpha_r(\mathbf{x}_i) = 1$ for all regions in this case. In our experiments, we split each image into several regions by same-size patches, and leverage (7) to mark the importance of different patches (regions) in each sample and its corresponding feature maps. Therefore, in (5), $\mathbf{F}^t_{l_t,r}(\mathbf{x}_i) \in \mathbb{R}^{c_{l_t} \times h^p_{l_t} \times w^p_{l_t}}$ and $\widehat{\mathbf{F}^s_{l_t,r}(\mathbf{x}_i)} \in \mathbb{R}^{c_{l_t} \times h^p_{l_t} \times w^p_{l_t}}$ are used to represent the patches of the features of teacher and student, respectively. $h^p_{l_t} \times w^p_{l_t}$ represents the spatial size of each patch. In addition, $R = (h_{l_t}/h^p_{l_t}) \times (w_{l_t}/w^p_{l_t})$ in (5) indicates the number of patches.

D. Distillation by Exploiting Sample Importance

Besides investigating the guiding potential of different regions of features, exploring the guiding capacity of different samples in dataset is also crucial in knowledge transfer. As shown in Fig. 1(c), the prediction probability can represent the capacity of models, and the models show variational capacity on classifying the samples in different categories, hence the confidence level on inferring different samples is changeable, leading to diverse distillation guiding capacity on different samples. In this section, we introduce the sample importance to investigate the guiding capacity on different samples, and avoid the uniform distillation for all samples. Thus, (5) can be rewritten as follows:

$$\mathcal{L}_{\text{InsKD}} = \sum_{i=1}^{b} \lambda(\mathbf{x}_i) \cdot \mathcal{L}_{\text{RegKD}_i} \tag{8}$$

where $\lambda(\mathbf{x}_i)$ represents the importance of $\mathbf{x}_i$ in a mini-batch.

It is a common practice that the teacher always imparts more assured knowledge to students. Similarly, we conduct a more powerful transfer for the samples that have higher confidence levels in the teacher model. The loss of teacher can be leveraged to describe the confidence level of samples, and then, the sample importance is calculated as follows:

$$\lambda(\mathbf{x}_i) = b \cdot \frac{\exp(-\mathcal{L}_{ce}(f^t(\mathbf{x}_i), \mathbf{y}_i)/\tau)}{\sum_{i'=1}^{b} \exp(-\mathcal{L}_{ce}(f^t(\mathbf{x}_{i'}), \mathbf{y}_{i'})/\tau)} \tag{9}$$

where $\mathcal{L}_{ce}(f^t(\mathbf{x}_i), \mathbf{y}_i)$ represents the loss of $\mathbf{x}_i$ in teacher network and $f^t(\mathbf{x}_i)$ represents the output of teacher model. τ is a factor to modulate the distribution of sample importance. A larger τ will get more flatter sample importance in a mini-batch. In (9), when the sample has higher confidence level, its prediction result and one-hot label are more similar, and then, $\mathcal{L}_{ce}(f^t(\mathbf{x}_i), \mathbf{y}_i)$ tends to be smaller, leading to a larger $\lambda(\mathbf{x}_i)$. This makes sample $\mathbf{x}_i$ play an important role in student learning. Conversely, the sample with a lower confidence level leads to smaller $\lambda(\mathbf{x}_i)$.

E. Final Loss of DenseKD

We combine the task loss and distillation loss to train the higher performance compact student models. The final loss function can be formulated as follows:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{task}} + \beta \cdot \mathcal{L}_{\text{DenseKD}} \\ &= \mathcal{L}_{\text{task}} + \beta \cdot \sum_{i=1}^{b} \lambda(i) \cdot \sum_{l_t=1}^{L_t} \mathcal{D}\big(\mathbf{MF}^t_{l_t}(i), \mathbf{MF}^s_{l_t}(i)\big)\end{aligned} \tag{10}$$

where $\mathbf{x}_i$ is simplified to i. $\mathbf{MF}^t_{l_t}(i) = \mathbf{M}_{l_t}(i) \odot \mathbf{F}^t_{l_t}(i)$, and $\mathbf{MF}^s_{l_t}(i) = \mathbf{M}_{l_t}(i) \odot \widehat{\mathbf{F}^s_{l_t}(i)}$. $\odot$ represents elementwise multiplication. $\mathbf{M}_{l_t}(i) \in \mathbb{R}^{c_{l_t} \times h_{l_t} \times w_{l_t}}$ represents the mask generated by the region importance $\alpha(i)$, where the value of each pixel is its corresponding region importance. $\mathcal{D}$ is mean square error in this article. β is used to balance two individual loss terms. It should be mentioned that although the dense connection in DenseKD increases the training time, the distillation loss is simply added alone with task loss during the student training process, and the inference is exactly the same as the original student model. Hence, our method is totally *cost-free at inference phase*.

IV. Experiment

A. Experimental Settings

1) Datasets and Networks: In the experiments, we apply DenseKD to compress the models on multiple vision tasks, including, classification and other high-level tasks (object detection and instance segmentation). In this subsection, according to different tasks, we briefly introduce the datasets and networks involved in the experiments.

TABLE I
Configurations for Experiments on CIFAR-100. "-" Indicates That the Basic Dense Connection Is Leveraged to Conduct the CKD

Distillation pair	β	m	Patch size	τ
ResNet-56 & ResNet-18	1.8	-	8×8	0.75
ResNet-110 & ResNet-32	3.0	-	8×8	0.25
ResNet32x4 & ResNet8x4	30.0	-	8×8	0.75
WRN40-2 & WRN16-2	7.0	2	8×8	0.25
WRN40-2 & WRN40-1	40.0	-	8×8	0.25
VGG-13 & VGG-8	100.0	4	8×8	0.25
ResNet32x4 & ShuffleNetV1	80.0	4	8×8	0.25
WRN40-2 & ShuffleNetV1	100.0	4	8×8	1.00
VGG-13 & MobileNetV2	40.0	4	8×8	1.00
ResNet50 & MobileNetV2	40.0	2	8×8	1.00
ResNet32x4 & ShuffleNetV2	100.0	4	8×8	0.25

a) Classification: We deploy DenseKD on two different benchmarks, including, CIFAR-100 [51] and ImageNet [52]. CIFAR-100 contains 60k images with the size of 32×32 . These images are evenly assigned to the 100 categories, where training set and validation set contain 50k and 10k images, respectively. In addition, ImageNet-1k is a large-scale dataset with 1k categories, and it contains 1.28 M training images and 50k validation images. The images are preprocessed to the size of 224×224 for the subsequent training and inference.

To illustrate that DenseKD has the ability to compress the DNNs on classification tasks, multiple architectures, involving, VGG [53] with the plain structure, ResNet [54] and Wide ResNet (WRN) [55] with the residual branch, MobileNet [56] with depthwise separable convolution, and ShuffleNet [57] with channel shuffle operation, are used to build homogeneous (e.g., "ResNet-56 and ResNet-20") and heterogeneous(e.g., "ResNet32 $\times$ 4 and ShuffleNetV1") teacher–student pairs.

b) Object detection and instance segmentation: COCO2017 [58] is used to illustrate the effectiveness of our method on high-level tasks. The COCO dataset has 118k images for training and 5k images for validation. All the objects in these images can be clustered into 80 categories, where approximately 41% are small objects, 34% for medium, as well as 24% for large.

In the experiments for high-level tasks, we compress both one-stage (RetinaNet [4]) and two-stage detectors (Faster R-CNN [59]) for detection, where the backbone of Faster R-CNN is built by ResNet and MobileNet. Furthermore, we compress Mask R-CNN with same backbones to show that DenseKD can be applied for instance segmentation task. It should be noted that the transferred features are extracted from the feature pyramid network (FPN) [60].

2) Configurations and Implementations: In this item, we introduce the experiment configurations for classification and high-level tasks, respectively.

a) Classification: For the experiments on CIFAR-100, the number of epochs, batch size, weight decay, and momentum are set to 240, 64, 5e−4, and 0.9, respectively. When MobileNet and ShuffleNet are student models, the initial learning rate is set to 0.01, and it is divided by 10 at 150, 180, and 210 epochs, respectively. Besides, the initial learning rate for other models is set to 0.05 with the same decaying scheme. The configurations of distillation are indicated in Table I.

For the experiments on ImageNet, we initialize the learning rate as 0.1 and decay it by 0.1 at 30, 60, and 90 epochs until the

TABLE II

RESULTS ON CIFAR-100. THE TEACHER AND STUDENT HAVE HOMOGENEOUS ARCHITECTURES. THE ACCURACY OF EACH METHOD AGAINST VANILLA KD [21] IS REPORTED IN THE PARENTHESES, WHERE RED AND BLUE MARK THE HIGHER AND LOWER ACCURACIES THAN VANILLA KD, RESPECTIVELY

Knowledge sources	Teacher Accuracy	ResNet-56 72.34	ResNet-110 74.31	ResNet32x4 79.42	WRN40-2 75.61	WRN40-2 75.61	VGG-13 74.64
	Student Accuracy	ResNet-20 69.09	ResNet-32 71.14	ResNet8x4 72.50	WRN16-2 73.26	WRN40-1 71.98	VGG-8 70.36
Logit	KD [21]	70.66	73.08	73.33	74.92	73.54	72.98
	DKD [62]	71.97 (+1.31)	74.11 (+1.03)	76.32 (+2.99)	76.24 (+1.32)	74.81 (+1.27)	74.68 (+1.70)
	Z-score + KD [63]	71.43 (+0.77)	74.17 (+1.09)	76.62 (+3.29)	76.11 (+1.19)	74.37 (+0.83)	74.36 (+1.38)
SKD	FitNet [26]	69.21 (-1.45)	71.06 (-2.02)	73.50 (+0.17)	73.58 (-1.34)	72.24 (-1.30)	71.02 (-1.96)
	PKT [64]	70.34 (-0.32)	72.61 (-0.47)	73.64 (+0.31)	74.54 (-0.38)	73.53 (-0.01)	72.88 (-0.10)
	RKD [65]	69.61 (-1.05)	71.82 (-1.26)	71.90 (-1.43)	73.35 (-1.57)	72.22 (-1.32)	71.48 (-1.50)
	CRD [34]	71.16 (+0.50)	73.48 (+0.40)	75.51 (+2.18)	75.48 (+0.56)	74.14 (+0.60)	73.94 (+0.96)
	CAT [35]	71.62 (+0.96)	73.62 (+0.54)	76.91 (+3.58)	75.60 (+0.68)	74.82 (+1.28)	74.65 (+1.67)
MKD	AT [28]	70.55 (-0.11)	72.31 (-0.77)	73.44 (+0.11)	74.08 (-0.84)	72.77 (-0.77)	71.43 (-1.55)
	VID [37]	70.38 (-0.28)	72.61 (-0.47)	73.09 (-0.24)	74.11 (-0.81)	73.30 (-0.24)	71.23 (-1.75)
	OFD [38]	70.98 (+0.32)	73.23 (+0.15)	74.95 (+1.62)	75.24 (+0.32)	74.33 (+0.79)	73.95 (+0.97)
CKD	SemCKD [29]	71.86 (+1.20)	73.94 (+0.86)	76.23 (+2.90)	75.78 (+0.86)	74.41 (+0.87)	74.43 (+1.45)
	ReviewKD [31]	71.89 (+1.23)	73.89 (+0.81)	75.63 (+2.30)	76.12 (+1.20)	75.09 (+1.55)	74.84 (+1.86)
	DenseKD	**72.30** (+1.64)	**74.73** (+1.65)	**76.47** (+3.14)	**76.49** (+1.57)	**75.54** (+2.00)	**75.17** (+2.19)

last 100 epochs. The batch size, weight decay, and momentum are set to 256, 1e−4, and 0.9, respectively. In addition, m is set to 4. τ and R are set to 0.25 and 9, respectively. It is worth mentioning that the deeper features obviously have too small spatial size, leading to ambiguous region importance, thus (10) is leveraged to investigate the region importance of the first two stages, i.e., if $l_t > 2$, then $\mathbf{M}_{l_t} = \mathbf{1}$.

b) Object detection and instance segmentation: we leverage Detectron2 [61] to deploy our method. It also provides the most popular pretrained teacher models for our experiments. We take the same training policy with [31] and [41] to train the compressed student model. For all the teacher–student combinations in one- and two-stage detectors, we set m to 2. Besides, the key of object detection is to select the objects (foreground) from the background and identify the local object instances in each global image, rather than only focusing on the foreground and classifying the entire image in the classification task [40]. Hence, all the regions in the backbone and all the samples are important, and then, we set α and λ to 1 for all the regions and samples, respectively, in the detection task.

3) Evaluation Metrics: For the experiments on classification, we report Top-1 accuracy for all the compressed models. In addition, Top-5 accuracy is an additional metric for the models on ImageNet. In addition, for the experiments on COCO2017, mean average precision (mAP), AP50, and AP70, as well as the metrics for evaluating the objects with different sizes (APl, APm, and APs) are adopted to illustrate the usefulness of DenseKD. All the models are trained for three times and the mean accuracy is reported in this article. All the experiments are implemented with Pytorch on Nvidia Titan XP and GeForce RTX 3090 GPUs.

B. Results on Classification

1) Results on CIFAR-100: In Table II, we display the distillation results of homogeneous teacher–student pairs on CIFAR-100 and compare the result of DenseKD with that of other KD methods, including, logit-based distillation (vanilla KD [21], DKD [62], Z-score [63], and SDD [66]), single-layer KD (FitNet [26], PKT [64], RKD [65], CRD [34], and CAT [35]), MKD (AT [28], VID [37], OFD [38]), and CKD (SemCKD [29] and ReviewKD [31]). As shown in Table II, DenseKD provides excellent Top-1 accuracy. For example, compared with the vanilla KD, DenseKD gains a ResNet-20 with 72.30% accuracy, which is much higher than 70.66% by vanilla KD and 69.06% of the raw student network. Also, for "ResNet32 × 4 and ResNet8 × 4," DenseKD improves the vanilla KD by more than 3%, although it is slightly lower than CAT. Besides, compared with SemCKD, DenseKD uses WRN40-2 to get a WRN40-1 with higher accuracy (75.51% versus 74.41%). Moreover, for the "ResNet-110 and ResNet-32" combination, DenseKD is the only one that outperforms teacher network among all the compared methods (74.73% versus 74.31%). Hence, the proposed method has ability to conduct distillation on homogeneous combinations.

In addition, Table III records the results of heterogeneous teacher-student pairs. DenseKD again surpasses most of its counterparts by a large margin. For example, compared with CKD methods (SemCKD and ReviewKD), DenseKD further improves the accuracy of student models (e.g., 78.09% versus 76.31% by SemCKD and 77.45% by ReviewKD on "ResNet32 × 4 and ShuffleNetV1"). Also, when leveraging WRN40-2 to supervise the ShuffleNetV1, our method also achieves the student model with the higher accuracy (e.g., 77.76% versus 77.11% by SemCKD, 77.14% by ReviewKD, and 75.85% by OFD). Therefore, DenseKD is effective for the teacher and student in combination with different architectures.

2) Results on ImageNet: To illustrate that DenseKD can compress the DNNs on complex datasets, we apply it to transfer the knowledge from ResNet-34 to ResNet-18 on ImageNet, and compare the result of DenseKD with that of other advanced methods, including, vanilla KD [21], AT [28],

TABLE III
RESULTS ON CIFAR-100. THE TEACHER AND STUDENT HAVE HETEROGENEOUS ARCHITECTURES

Knowledge sources	Teacher Accuracy	ResNet32x4 79.42	WRN40-2 75.61	VGG-13 74.64	ResNet-50 79.34	ResNet32x4 79.42
	Student Accuracy	ShuffleNetV1 70.50	ShuffleNetV1 70.50	MobileNetV2 64.60	MobileNetV2 64.60	ShuffleNetV2 71.82
Logit	KD [21]	74.07	74.83	67.37	67.35	74.45
	DKD [62]	76.45 (+2.38)	76.70 (+1.87)	69.71 (+2.34)	70.35 (+3.00)	77.07 (+2.62)
	Z-score + KD [63]	-	-	68.61 (+1.24)	69.02 (+1.67)	75.56 (+1.11)
	SDD + KD [66]	76.30 (+2.23)	76.65 (+1.82)	68.79 (+1.42)	69.55 (+2.20)	76.67 (+2.22)
SKD	FitNet [26]	73.59 (-0.48)	73.73 (-1.10)	64.14 (-3.23)	63.16 (-4.19)	73.54 (-0.91)
	PKT [64]	74.10 (+0.03)	73.89 (-0.94)	67.13 (-0.24)	66.52 (-0.83)	74.69 (+0.24)
	RKD [65]	72.28 (-1.79)	72.21 (-2.62)	64.52 (-2.85)	64.43 (-2.92)	73.21 (-1.24)
	CRD [34]	75.11 (+1.04)	76.05 (+1.22)	69.73 (+2.36)	69.11 (+1.76)	75.65 (+1.20)
	CAT [35]	78.26 (+4.18)	77.35 (+2.52)	69.13 (+1.76)	71.36 (+4.01)	78.41 (+3.96)
MKD	AT [28]	71.73 (-2.34)	73.32 (-1.51)	59.40 (-7.97)	58.58 (-8.77)	72.73 (-1.72)
	VID [37]	73.38 (-0.69)	73.61 (-1.22)	65.56 (-1.81)	67.57 (+0.22)	73.40 (-1.05)
	OFD [38]	75.98 (+1.91)	75.85 (+1.02)	69.48 (+2.11)	69.04 (+1.69)	76.82 (+2.37)
CKD	SemCKD [29]	76.31 (+2.24)	77.11 (+2.28)	68.41 (+1.04)	-	77.62 (+3.17)
	ReviewKD [31]	77.45 (+3.38)	77.14 (+2.31)	70.37 (+3.00)	69.89 (+2.54)	77.78 (+3.33)
	DenseKD	**78.09** (+4.02)	**77.76** (+2.93)	**71.22** (+3.85)	**70.55** (+3.20)	**78.14** (+3.69)

CAT [35], OFD [38], CRD [34], CC [67], RKD [65], IRG [68], and SP [69]. As shown in Table IV, Our method outperforms other distillation methods, and it increases the Top-1 and Top-5 accuracies by 1.89% and 1.41% over the original student model, respectively. Therefore, DenseKD also demonstrates its potential to conduct knowledge transfer on large-scale datasets.

C. Results on Object Detection

To demonstrate that the proposed method is effective on other vision tasks, we utilize DenseKD to transfer the knowledge of one-stage (RetinaNet) and two-stage (Faster R-CNN) detectors trained on COCO2017, respectively. As shown in Table V, we compare the result of DenseKD with that of vanilla KD [21], DKD [62], FitNet [26], ReviewKD [31], and Wang et al. [41], where vanilla KD, DKD, FitNet, and ReviewKD are not the methods specifically designed for object detection task, while Wang et al. [41] is a detection-customized KD method. As shown in Table V, DenseKD achieves the preferable performance on object detection task. For the two-stage detector, when transferring the knowledge between backbones with the same type of architecture, DenseKD obtains the student models with higher mAP (e.g., 36.81% versus 33.97% by vanilla KD, 34.99% by DKD, 34.13% by FitNet, 36.21% by ReviewKD, and 35.44% by Wang et al. [41] for "ResNet-18 and ResNet-110"). Besides, DenseKD also brings powerful capacity for the combination with heterogeneous backbones. For example, compared with the student trained without distillation loss, the proposed method greatly improves the detection capacity of the student model (33.72% versus 29.47%, 53.38% versus 48.87%, and 36.02% versus 30.90% on mAP, AP50, and AP75, respectively). Moreover, DenseKD is also effective for RetinaNet. For instance, compared with Wang et al. [41], our method boosts the mAP of the compact model by 1.34%.

It is worth mentioning that DenseKD slightly improves the results of ReviewKD in a few cases (e.g., 33.72% versus 33.51% on mAP for "ResNet-50 and MobileNetV2"), as it only updates the distillation strategy for the backbone. As illustrated in [39], [40], [41], [43], and [45], investigating the balance of the positive and negative instances, and exploring the knowledge of other components (e.g., detection head) are critical for conducting KD on detection task. To further explain the potential of DenseKD on detection task, we will combine it with additional loss in the ablation study (Section V-J).

D. Results on Instance Segmentation

In addition, we further conduct our method on more challenging instance segmentation task, and Mask R-CNN with different backbone on COCO2017 is compressed. As shown in Table VI, when ResNet-101 is set to the backbone of teacher network, DenseKD still gets lightweight Mask R-CNN with higher mAP (33.51% versus 31.24% for ResNet18-based model, and 37.02% versus 35.24% for ResNet50-based one). Also, for the combination of "ResNet-50 and MobileNetV2," DenseKD also improves 2.49% of mAP of the student model when compared with training without KD loss.

To sum up, the remarkable distillation results on multiple combinations indicate that DenseKD has great capacity to transfer the knowledge by building dense connections, investigating the guiding capacity of different regions by region importance, and exploring the guiding potential of samples by sample importance. Although DenseKD is mainly proposed to compress the model on classification tasks, it can also be generalized to other high-level vision tasks. In the next section, we further verify the effectiveness of each component in DenseKD through ablation studies.

V. ABLATION STUDY

A. Visualization of Selected Channels

We leverage the combinations of "VGG-13 and VGG-8" and "VGG-13 and MobileNetV2" as examples to visualize the guiding of multiple-stage teacher features on the channel-wise features of each student stage. The configurations are the same as those in Section IV-A. Since the magnitude of element in convolution can indicate the capacity of convolution [70], [71], [72], we calculate the norm of refining module corresponding

TABLE IV
RESULTS ON IMAGENET. THE TEACHER NETWORK IS RESNET-34 AND THE STUDENT NETWORK IS RESNET-18

	Teacher	Student	KD [21]	AT [28]	CAT [35]	OFD [38]	CRD [34]	CC [67]	RKD [65]	IRG [68]	SP [69]	**DenseKD**
Top-1 accuracy	73.31	69.75	70.66	70.69 (+0.03)	71.26 (+0.60)	70.81 (+0.15)	71.17 (+0.51)	69.96 (-0.70)	70.59 (-0.07)	70.32 (-0.34)	70.79 (+0.13)	**71.64** (+0.98)
Top-5 accuracy	91.42	89.07	89.88	90.01 (+0.13)	90.45 (+0.57)	89.98 (+0.10)	90.13 (+0.25)	89.17 (-0.71)	89.68 (-0.20)	89.99 (+0.11)	89.80 (-0.08)	**90.48** (+0.60)

TABLE V
RESULTS ON OBJECT DETECTION. THE EVALUATIONS OF EACH METHOD AGAINST STUDENT MODEL ARE REPORTED IN THE PARENTHESES. "*" MARKS RE-IMPLEMENTED RESULTS BASED ON THE RELEASED CODE

	Method	mAP	AP50	AP75	APl	APm	APs
Teacher	Faster R-CNN w. ResNet-101	42.04	62.48	45.88	54.60	45.55	25.22
Student	Faster R-CNN w. ResNet-18	33.26	53.61	35.26	43.16	35.68	18.96
	+ vanilla KD [21]	33.97 (+0.71)	54.66 (+1.05)	36.62 (+1.36)	44.14 (+0.98)	36.67 (+0.99)	18.71 (-0.25)
	+ DKD* [62]	34.99 (+1.73)	56.49 (+2.88)	37.19 (+1.93)	46.01 (+2.85)	37.60 (+1.92)	**19.72** (+0.76)
	+ FitNet [26]	34.13 (+0.87)	54.16 (+0.55)	36.71 (+1.45)	44.69 (+1.53)	36.50 (+0.82)	18.88 (-0.08)
	+ ReviewKD* [31]	36.21 (+2.95)	56.28 (+2.67)	38.84 (+3.58)	49.06 (+5.90)	38.81 (+3.13)	19.20 (+0.24)
	+ Wang et al. [41]	35.44 (+2.18)	55.51 (+1.90)	38.17 (+2.91)	47.34 (+4.18)	38.29 (+2.61)	19.04 (+0.08)
	+ DenseKD	**36.81** (+3.55)	**56.60** (+2.99)	**39.89** (+4.63)	**49.69** (+6.53)	**39.51** (+3.83)	19.29 (+0.33)
Teacher	Faster R-CNN w. ResNet-101	42.04	62.48	45.88	54.60	45.55	25.22
Student	Faster R-CNN w. ResNet-50	37.93	58.84	41.05	49.10	41.14	22.44
	+ vanilla KD [21]	38.35 (+0.42)	59.41 (+0.57)	41.71 (+0.66)	49.48 (+0.38)	41.80 (+0.66)	22.73 (+0.29)
	+ DKD* [62]	39.42 (+1.49)	61.02 (+2.18)	42.71 (+1.66)	51.48 (+2.38)	42.91 (+1.77)	**23.54** (+1.10)
	+ FitNet [26]	38.76 (+0.83)	59.62 (+0.78)	41.80 (+0.75)	50.70 (+1.60)	42.20 (+1.06)	22.32 (-0.12)
	+ ReviewKD* [31]	40.13 (+2.20)	60.66 (+1.82)	43.76 (+2.71)	52.62 (+3.52)	43.78 (+2.64)	22.96 (+0.52)
	+ Wang et al. [41]	39.44 (+1.51)	60.27 (+1.43)	43.04 (+1.99)	51.97 (+2.87)	42.51 (+1.37)	22.89 (+0.45)
	+ DenseKD	**40.91** (+2.98)	**61.39** (+2.55)	**44.77** (+3.72)	**53.58** (+4.48)	**44.59** (+3.45)	23.31 (+0.87)
Teacher	Faster R-CNN w. ResNet-50	40.22	61.02	43.81	51.98	43.53	24.16
Student	Faster R-CNN w. MobileNetV2	29.47	48.87	30.90	38.86	30.77	16.33
	+ vanilla KD [21]	30.13 (+0.66)	50.28 (+1.41)	31.35 (+0.45)	39.56 (+0.70)	31.91 (+1.14)	16.69 (+0.36)
	+ DKD* [62]	32.19 (+2.72)	53.65 (+4.78)	33.84 (+2.94)	42.42 (+3.56)	34.39 (+3.62)	**19.02** (+2.69)
	+ FitNet [26]	30.20 (+0.73)	49.80 (+0.93)	31.69 (+0.79)	39.69 (+0.83)	31.64 (+0.87)	16.39 (+0.06)
	+ ReviewKD* [31]	33.51 (+4.04)	53.16 (+4.29)	35.93 (+5.03)	45.30 (+6.44)	35.86 (+5.09)	18.04 (+1.71)
	+ Wang et al. [41]	31.16 (+1.69)	50.68 (+1.81)	32.92 (+2.02)	42.12 (+3.26)	32.63 (+1.86)	16.73 (+0.40)
	+ DenseKD	**33.72** (+4.25)	**53.38** (+4.51)	**36.02** (+5.12)	**45.41** (+6.55)	**36.11** (+5.34)	17.32 (+0.99)
Teacher	RetinaNet-101	40.40	60.25	43.19	52.18	44.34	24.03
Student	RetinaNet-50	36.15	56.03	38.73	46.95	40.25	21.37
	+ vanilla KD [21]	36.76 (+0.61)	56.60 (+0.57)	39.40 (+0.67)	48.17 (+1.22)	40.56 (+0.31)	21.87 (+0.50)
	+ FitNet [26]	36.30 (+0.15)	55.95 (-0.08)	38.95 (+0.22)	47.14 (+0.19)	40.32 (+0.07)	20.10 (-1.27)
	+ ReviewKD* [31]	38.48 (+2.33)	58.22 (+2.19)	41.46 (+2.73)	**51.15** (+4.20)	42.72 (+2.47)	22.67 (+1.30)
	+ Wang et al. [41]	37.29 (+1.14)	57.13 (+1.10)	40.04 (+1.31)	49.71 (+2.76)	41.47 (+1.22)	21.01 (-0.36)
	+ DenseKD	**38.63** (+2.48)	**58.75** (+2.72)	**41.73** (+3.00)	50.06 (+3.11)	**43.11** (+2.86)	**22.82** (+1.45)

to each channel of the feature map to illustrate the influence of teacher on student. It can be formulated as follows:

$$CI_{l_s,l_t}(c_1) = \sum_{c_2=1}^{c_{l_s}/m} \left|\mathbf{w}_{l_s,l_t}(c_1, c_2, :, :)\right| \tag{11}$$

where $\mathbf{w}_{l_s,l_t} \in \mathbb{R}^{c_{l_s} \times c_{l_s \backslash m} \times 1 \times 1}$ represents the refining module between the l_sth student features and the l_tth teacher feature, $c_1 \in [1, c_{l_s}]$, $CI_{l_s,l_t}(c_1)$ represents the influence of l_tth teacher features on the c_1th channel of l_sth student features.

As shown in Fig. 5, each figure records the influence of multistage features of the teacher model on each-stage features of the student model. According to the visualization results, we get the following rules.

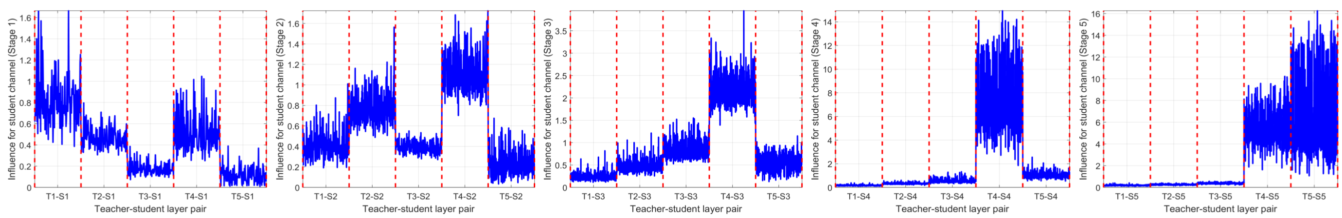
1) Since the blue curves fluctuate significantly, the features of each teacher stage conduct diverse supervision on different channelwise features of each student stage. Therefore, different from stage-wise feature matching in [29] and [31], it is necessary to select the fine-grained student channels to capture the teacher features.
2) Not all the features of the previous stage can supervise the current features of the student, this is different from review mechanism [31], which uses shallower features to guide the current features. For example, in the rightmost of Fig. 5(a) and (b), the features at 1st, 2nd, and 3rd teacher stages (shallower features) hardly provide guiding for the features at 5th student stage.
3) Not only the features of current and previous teacher layers can guide the learning of the current student layer but also the features of later layers have the ability to provide supervision. For example, in the third figure of Fig. 5(a), the features of fourth stage of VGG-13 also transfer to the third stage of VGG-8.

In summary, the knowledge of the teacher has different effects on the channel of each student stage; thus, it is valuable to conduct a CKD in a fine-grained and adaptive fashion.

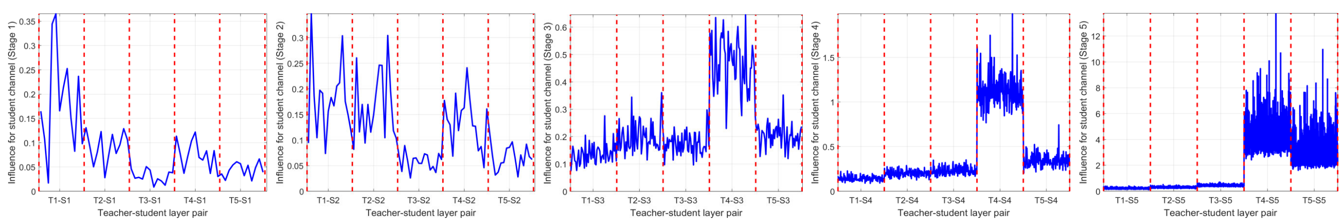
TABLE VI

RESULTS ON INSTANCE SEGMENTATION. THE EVALUATIONS OF EACH METHOD AGAINST STUDENT MODEL ARE REPORTED IN THE PARENTHESES

	Method	mAP	AP50	AP75	APl	APm	APs
Teacher	Mask R-CNN w. ResNet-101	38.63	60.45	41.28	55.29	41.33	19.48
Student	Mask R-CNN w. ResNet-18	31.25	51.07	33.10	45.53	32.80	14.18
	+ **DenseKD**	**33.51** (+2.26)	**53.76** (+2.69)	**35.67** (+2.57)	**50.40** (+4.87)	**35.18** (+2.28)	**14.58** (+0.40)
	Method	mAP	AP50	AP75	APl	APm	APs
Teacher	Mask R-CNN w. ResNet-101	38.63	60.45	41.28	55.29	41.33	19.48
Student	Mask R-CNN w. ResNet-50	35.24	56.32	37.49	50.34	37.71	17.16
	+ **DenseKD**	**37.02** (+1.78)	**58.19** (+1.87)	**39.73** (+2.24)	**53.69** (+3.35)	**39.63** (+1.92)	**17.33** (+0.17)
	Method	mAP	AP50	AP75	APl	APm	APs
Teacher	Mask R-CNN w. ResNet-50	37.17	58.60	39.88	53.30	39.49	18.63
Student	Mask R-CNN w. MobileNetV2	28.37	47.19	29.95	41.70	29.01	12.09
	+ **DenseKD**	**30.86** (+2.49)	**50.37** (+3.18)	**32.77** (+2.82)	**46.18** (+4.48)	**32.25** (+3.24)	**12.82** (+0.73)



(a)

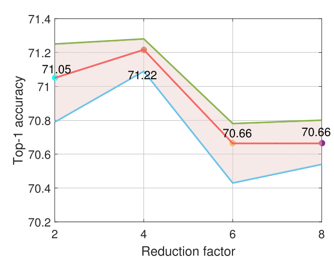


(b)

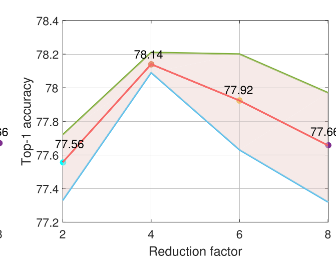
Fig. 5. Influence of multiple-stage teacher features on each-stage student features. x- and y-axes represent the teacher–student layer pair and CI_{l_s,l_t}, respectively. The blue curve between two red dotted lines represents the influence of one-stage teacher features on all the channelwise features at one student stage. For example, "T1–S1" in the leftmost represents the influence of the first teacher stage's features on the channelwise features of the first student stage. (a) VGG-13 to VGG-8. (b) VGG-13 to MobileNetV2.

B. Influence of Reduction Factor

In Section III-B, the fast dense connection constructs the pathway for knowledge transfer, where reduction factor (m) will affect the number of outputs of the refining module, and then affects the efficiency of knowledge transfer. A larger m means a faster transfer. In this section, we take the combinations of "VGG-13 and MobileNetV2" and "ResNet32 × 4 and ShuffleNetV2" as examples to illustrate the influence of reduction factor. For each combination, we set m to 2, 4, 6, and 8, other settings are the same as those in Section IV-A. As shown in Fig. 6, when m is particularly large, knowledge transfer becomes faster, however, the student network accuracy becomes lower as the parameters are reduced. For example, the accuracy of MobileNetV2 when $m = 6$ and 8 is lower than that of MobileNetV2 when $m = 4$ (70.66% versus 71.22%). Moreover, fewer parameters may also provide the high-accuracy model. For instance, compared with the accuracy of ShuffleNetV2 at $m = 2$ and $m = 4$ leads to higher performance. This may be because, at $m = 4$, the number of channels in each stage is optimal (neither redundant nor insufficient), allowing each stage to obtain knowledge in a more balanced manner. Therefore, the appropriate reduction factor should be selected to balance the accuracy of student model and distillation efficiency.



(a)



(b)

Fig. 6. Influence of reduction factor. x- and y-axes represent the reduction factor and accuracy, respectively. For each combination, we conducted experiments several times. The accuracy varies between the blue and green curves, and the red curve represents the mean accuracy. (a) VGG-13 and MobileNetV2. (b) ResNet32 × 4 and ShuffleNetV2.

C. Influence of Patch Size

In Section III-C, the images and feature maps are spilt into multiple patches to investigate region importance

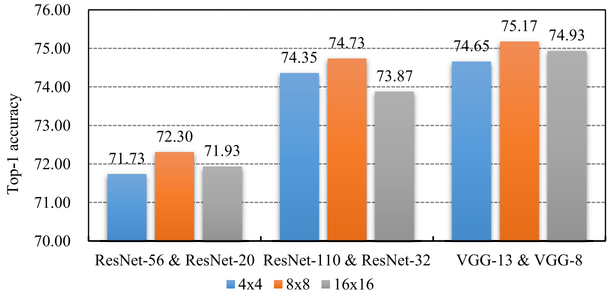


Fig. 7. Influence of patch size. The blue, orange, and gray bars represent the results of patch size with "4 × 4," "8 × 8," and "16 × 16," respectively. Each set of three bars is the results of one distillation pair. x- and y-axes represent the distillation pair and the accuracy of the student model, respectively.

in distillation. In this section, we take "ResNet-56 and ResNet-20," "ResNet-110 and ResNet-32," and "VGG-13 and VGG-8" combinations as examples to demonstrate the influence of patch size. We set the patch size to 4 × 4, 8 × 8, and 16 × 16, respectively, other hyperparameters are introduced in Section IV-A. As shown in Fig. 7, DenseKD brings the optimal student models when patch size is 8 × 8 (72.30%, 74.73%, and 75.17% for three combinations, respectively) and both larger and smaller patch size produce the suboptimal network. The possible explanations are as follows: the importance of region with the least important is set to be 0 by (7); hence, large region is ignored in distillation when setting the excessively large patch size, resulting in the accuracy drop of student model. In addition, when the patch size is excessively small, the target content contained in each patch is not enough to produce the different $\mathcal{D}_{\text{reg}}(\mathbf{x}_{i,r})$ for each region, leading to similar region importance for different regions, which then affects the results. Also, it should be noted that the finer grid brings more training costs. Therefore, the moderate patch size achieves networks with higher performance.

D. Influence of Variant Region Importance

The nonoverlapped region is leveraged to calculate the region importance in our method. To further illustrate the effectiveness of this scheme, we compare the distillation results of this scheme and its variants. In the variant scheme, we calculate the region importance with the overlapped region, where the sidelength of the patch is set to 16 and 24, respectively. Also, for each sidelength, the stride is set to 4 and 8, respectively. Since the importance of the overlapped region is calculated multiple times in different patches, we take the average of the results of multiple times as the region importance of the overlapped region.

Table VII demonstrates the results on different distillation pairs. The last column presents the results of calculating region importance with nonoverlapped region (our method), while other columns display the results of the overlapped-region-based scheme. In the table, calculating region importance with overlapped and nonoverlapped regions can get similar distillation results. For example, for "ResNet-32 × 4 and ShuffleNetV2," both schemes yield the student model with the accuracy of 75.14%, where the patch size and stride are set to 24 × 24 and 8, respectively, for the method based on overlapped region. Therefore, calculating region importance with either overlapped regions or nonoverlapped regions can get accurate compact student models.

TABLE VII

Influence of Variant Region Importance Scheme. The Patches With Different Sizes and Strides Are Used to Calculate Region Importance. In the Configurations of Patch, the First and Second Values in the Bracket Indicate the Sidelength and Stride of the Patch, Respectively

Distillation pair	Configurations of patch (h^p, s^p)				
	(16, 4)	(16, 8)	(24, 4)	(24, 8)	(8, 8)
ResNet56 & ResNet20	72.02	71.96	71.90	72.01	**72.30**
ResNet110 & ResNet32	74.38	74.68	74.27	73.92	**74.73**
ResNet32x4 & ShuffleNetV1	77.99	77.75	78.08	77.65	**78.09**
ResNet32x4 & ShuffleNetV2	77.93	77.64	77.94	**78.14**	**78.14**

E. Visualization of Region Importance

In this section, we visualize the importance of different regions of images to illustrate the effectiveness of region importance. We randomly select several images from ImageNet, and then ResNet-34 is taken as the teacher model to mark the importance of different regions by the method introduced in Section III-D, where each image is split into nine nonoverlapping patches. As shown in Fig. 8, by investigating the region importance, the intuitive main target regions in images are marked as critical regions (red patches in figures), which encourages teacher to transfer more knowledge of target regions to student. For example, in the first three figures, birds are marked as primary regions, which will be mainly focused during distillation. However, the foliage and sky are transferred as secondary content.

F. Effectiveness of Sample Importance

In (9), we set the samples with higher confidences level to play more important role in distillation. To illustrate the effectiveness of this scheme (9), we introduce the reverse sample importance, and compare the results of different schemes. The reverse sample importance is formulated as follows:

$$\lambda_r(\mathbf{x}_i) = b \cdot \frac{\exp(\mathcal{L}_{ce}(f^t(\mathbf{x}_i), \mathbf{y}_i)/\tau)}{\sum_{i'=1}^{b} \exp(\mathcal{L}_{ce}(f^t(\mathbf{x}_{i'}), \mathbf{y}_{i'})/\tau)}. \tag{12}$$

Different from (9), in (12), the samples with higher confidence levels will obtain less attention in KD. For the fair comparison, we further unify $\lambda_r(\mathbf{x}_i)$ and $\lambda(\mathbf{x}_i)$ into the same range. Then, the reverse sample importance is rewritten as follows:

$$\lambda_r^*(\mathbf{x}_i) = \frac{\lambda_r(\mathbf{x}_i) - \min(\lambda_r(\mathbf{x}_i))}{\max(\lambda_r(\mathbf{x}_i)) - \min(\lambda_r(\mathbf{x}_i))} \cdot a_1 + a_2 \tag{13}$$

where $a_1 = \max(\lambda(\mathbf{x}_i)) - \min(\lambda(\mathbf{x}_i))$ and $a_2 = \min(\lambda(\mathbf{x}_i))$. $\max(\lambda(\mathbf{x}_i))$ and $\min(\lambda(\mathbf{x}_i))$ represent the maximum and minimum of $\lambda(\mathbf{x}_i)$, respectively.

As shown in Table VIII, we conduct experiments on different distillation pairs trained on the CIFAR-100 dataset, the results of our method are higher than those of the reverse sample importance scheme. For example, for "ResNet-56 and ResNet-20," our method achieves the ResNet-20 with an accuracy of 72.30%, which is higher than the result of reverse version (70.26%). Therefore, our approach, which focuses more on high-confidence samples, is effective.

G. Influence of Factor τ

In Section III-D, the factor τ is used to modulate the distribution of sample importance, a larger τ leads to the

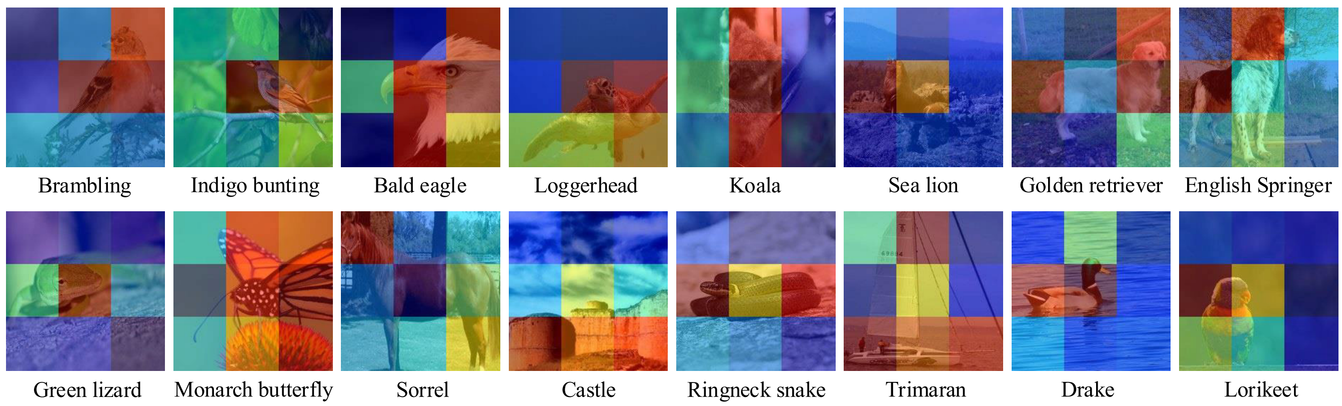


Fig. 8. Visualization of the importance of different regions. The ground-truth labels are demonstrated below each image. Regions with red are more important, while dark blue marks the minor regions. Best viewed in color.

TABLE VIII

COMPARISON OF DIFFERENT SAMPLE IMPORTANCE SCHEMES. "REVERSE" REPRESENTS THE REVERSE SAMPLE IMPORTANCE

Teacher	Student	Reverse	Ours
ResNet56	ResNet20	70.26	72.30 (+2.04)
ResNet110	ResNet32	73.06	74.73 (+1.67)
ResNet32x4	ShuffleNetV1	77.60	78.09 (+0.49)
ResNet32x4	ShuffleNetV2	77.71	78.14 (+0.43)

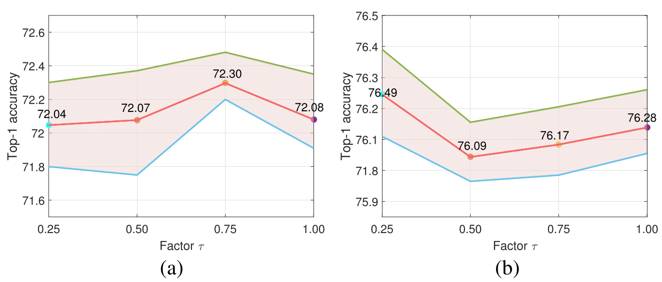


Fig. 9. Influence of Factor τ. x- and y-axes represent the factor τ and accuracy, respectively. For each combination, we conducted experiments several times. The accuracy varies between the blue and green curves, and the red curve represent the mean accuracy. (a) ResNet-56 and ResNet-20. (b) WRN40-2 and WRN16-2.

flatter distribution of sample importance. In this section, we take "ResNet-56 and ResNet-20" and "WRN40-2 and WRN16-2" as examples, and set τ to 0.25, 0.50, 0.75, and 1.00, respectively, to explain the influence of τ. Other configurations are stated in Section IV-A. As shown in Fig. 9, different combinations gain the optimal accuracy with different factors. For "ResNet-56 and ResNet-20," a relatively flat sample importance distribution ($\tau = 0.75$) can provide a more accurate student network (72.30%), while the importance with too flat or steep distributions will lead to suboptimal results. In addition, for "WRN40-2 and WRN16-2," a steeper sample importance distribution ($\tau = 0.25$) brings the best accuracy, which even higher than that of teacher network (76.49% versus 75.61%).

H. Influence of Different Components

As shown in Table IX, we further take "ResNet-56 and ResNet-20" combination as an example to illustrate the influence of each dense connection, region importance, and sample importance, respectively. All the configurations are the same as Section IV-A. According to Table IX, we can conclude the following items.

TABLE IX

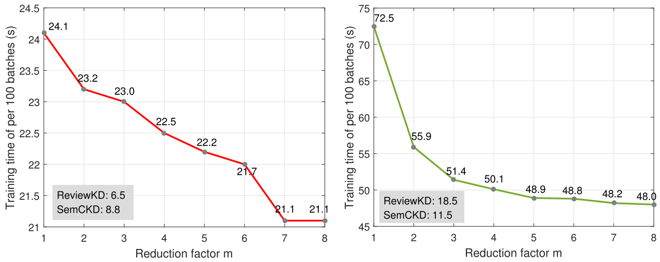
INFLUENCE OF DIFFERENT COMPONENTS. THE ACCURACY OF EACH SCHEME AGAINST TRAINING WITHOUT ANY COMPONENT IS REPORTED IN THE PARENTHESES

Dense connection	Region importance	sample importance	Accuracy
✘	✘	✘	71.06
✘	✔	✘	71.20 (+0.14)
✘	✘	✔	71.44 (+0.38)
✘	✔	✔	71.66 (+0.60)
✔	✘	✘	71.44 (+0.38)
✔	✔	✘	71.88 (+0.82)
✔	✘	✔	71.79 (+0.73)
✔	✔	✔	**72.30** (+1.24)

1) By comparing 1st and 5th row, 2nd and 6th row, 3rd and 7th row, and 4th and 8th row, respectively, the results of cross-layer distillation with the learnable and dense connection are significantly higher than those of multiple-layer distillation. Hence, the learnable dense connection can adaptively select the suitable channels of students to receive the knowledge from teacher.
2) In the first four rows, compared with the simple MKD in the first row, MKD with region importance (row 2) and MKD with sample importance (row 3) provide the models with higher accuracy, respectively. When both the region and sample importance are investigated, the accuracy of the model is further improved (row 4). In addition, comparing the last four rows yields the same conclusion.
3) When we aggregate all the components together, the student model achieves the highest accuracy. Therefore, dense connections, distillation with region, and sample importance can actively promote students to acquire knowledge from teachers.

I. Computational Overhead of Training Phase

Although DenseKD can obtain the student model with higher accuracy, the dense architecture and the additional teacher inference for calculating region importance sacrifice training time. In this section, to demonstrate the training



(a) (b)

Fig. 10. Training time of different distillation pair. x-axis represents the reduction factor (m), the y-axis represents the average training time per 100 batches. Batch size is set to 64, and an Nvidia Titan XP GPU is utilized. (a) WRN40-2 and WRN16-2. (b) ResNet32 $\times$ 4 and ShuffleNetV1.

resource consumption of DenseKD, we take "WRN40-2 and WRN16-2" and "ResNet32 $\times$ 4 and ShuffleNetV1" as examples to report the average training time per 100 batches of the training phase under different reduction factor (m). As shown in Fig. 10, as m increases, the number of parameters in the dense architecture decreases, thus reducing the training time. Moreover, ReviewKD [31] and SemCKD [29] require less training time because they transfer knowledge through modules with fewer parameters. It should be mentioned that although our method is more time-consuming in the training phase, in the inference phase, our method achieves the student models with higher accuracy and the same computational overhead as the models yielded by other methods. It is applicable to scenarios in which frequent training is not required and the computational resources and memory are limited. For example, embedded devices, such as cameras, drones, robotics, and autonomous driving systems have limited computational resources and require models capable of real-time inference. The compact models generated by DenseKD have the potential to conduct fast inference and are straightforward to deploy.

J. Extension of DenseKD

As stated in [39], [40], [41], [43], and [45], it is vital to balance the positive and negative instances for the deployment of KD on detection tasks. In this subsection, we combine the proposed method with the distillation loss of the classification term of detection head introduced in [39] to further extend DenseKD to the detection task. The overall loss is as follows:

$$\mathcal{L}_{\text{DenseKD+}} = \mathcal{L} + \sum_{i=1}^{N} \kappa_i \cdot \mathcal{D}_{\mathcal{KL}}\left(p_i^t || p_i^s\right) \tag{14}$$

where the first term is shown in (10), and the second term is the additional loss, which is utilized to investigate the knowledge of the detection head and balance the positive and negative instances. $\mathcal{D}_{\mathcal{KL}}$ represents the KL divergence. κ_i represents the weight of instance i, which balances the different instance. p_i^t and p_i^s represent the soften-logits of the detection heads of student and teacher, respectively. Note that, N represents the number of instances here. We demonstrate the results of Faster R-CNN with different backbones for explanation. The temperature of soften-logits is set to 1 [39], and other experiment settings are the same as those in Section IV-A. For each teacher–student pair, we set two combinations of the instance weights, including, "$\kappa_p = 1.0$ and $\kappa_n = 1.0$" and "$\kappa_p = 1.0$ and $\kappa_n = 2.0$", where κ_p and κ_n represent the weight of positive and negative instances, respectively. As shown in Table X, the extension of DenseKD (DenseKD+) achieves the compact student models with more outstanding detection ability. When the positive and negative instances have the same weights, DenseKD+ causes a few boosts, thus exploring more knowledge is beneficial to the application of DenseKD in the detection task. Furthermore, after balancing the different instances ($\kappa_p = 1.0$ and $\kappa_n = 2.0$), the student networks gains more significant improvements (e.g., 37.30%-mAP vs. 36.81%-mAP by DenseKD+ and DenseKD, respectively, for "ResNet-101 and ResNet-18"). Therefore, investigating the balance of positive and negative instances helps DenseKD achieve more excellent results in the detection task.

TABLE X

RESULTS ON DISTILLATION WITH EXTENSION OF DENSEKD. κ_p AND κ_n REPRESENT THE WEIGHT OF POSITIVE AND NEGATIVE INSTANCES, RESPECTIVELY

Backbone	Method	κ_p	κ_n	mAP	AP50	AP75
ResNet-101 & ResNet-18	DenseKD	-	-	36.81	56.60	39.89
	DenseKD+	1.0	1.0	36.91	57.38	40.11
		1.0	2.0	**37.30**	**57.88**	**40.30**
ResNet-50 & MobileNetV2	DenseKD	-	-	33.72	53.38	36.02
	DenseKD+	1.0	1.0	34.59	54.77	37.01
		1.0	2.0	**34.76**	**55.12**	**37.49**

In summary, DenseKD is effective for detection, even a slight gain is obtained against a few methods. Moreover, by introducing the detection-customized knowledge, DenseKD has ability to provide more superior student networks.

VI. CONCLUSION

This work proposes a fine-grained CKD by investigating the region and sample importance. To accurately transfer the knowledge, we first construct a learnable dense connection to select the distillation position in an adaptive and fine-grained scheme. In addition, we investigate the region's importance without relying on ground-truth and explore the sample importance in feature alignment to avoid uniform distillation for all regions and all samples, respectively. Extensive experiments on multiple vision tasks demonstrate that the proposed method achieves satisfactory results. Also, several ablation studies illustrate the effectiveness of each component.

Although the proposed method has the ability to achieve the compact student models with preferable performance, the imbalance of positive and negative instances in detection and segmentation tasks can cause performance limitations. In addition, DenseKD requires more computational overhead in the training phase compared with previous methods. Therefore, in the future work, we will focus on balancing positive and negative instances in detection and segmentation tasks, and improving the efficiency of knowledge transfer.

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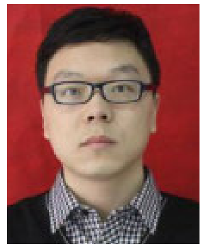

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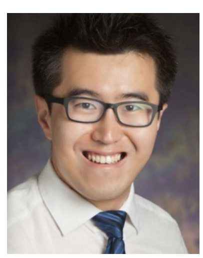

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